
Math 39

Precalculus with Data Analysis

Math 39 Course at a Glance

Precalculus with Data Analysis

Linear Modeling and Residuals

REQUIRED

M39-LIN-001	I can calculate and interpret average rate of change with appropriate units.
M39-LIN-002	I can construct a linear function from two data points, a table, a graph, or a context.
M39-LIN-003	I can fit a linear regression model to data using technology and interpret its parameters.
M39-LIN-004	I can interpret a correlation coefficient as evidence about the direction and strength of a linear relationship.
M39-LIN-005	I can calculate residuals, construct a residual plot, and use its pattern to evaluate a model.
M39-LIN-006	I can use a linear model to make and interpret predictions, including solving for either variable.

Exponential Modeling

REQUIRED

M39-EXP-001	I can distinguish linear and exponential change using rates, multipliers, tables, graphs, or contexts.
M39-EXP-002	I can construct an exponential function from two data points, a table, or a context.
M39-EXP-003	I can interpret the initial value, growth or decay factor, and growth or decay rate of an exponential model.
M39-EXP-004	I can determine and interpret doubling time or half-life from an exponential model.
M39-EXP-005	I can solve exponential and logarithmic equations in modeling contexts.
M39-EXP-006	I can fit an exponential regression model using technology and evaluate its fit.
M39-EXP-007	I can compare linear and exponential models and justify which is more appropriate for a data set or context.

Power Functions and Function Comparison

REQUIRED

M39-POW-001	I can identify a power function and interpret the parameters in $y = kx^p$.
M39-POW-002	I can construct a power function from two data points or a proportional relationship.
M39-POW-003	I can relate the sign and size of k and p to the shape, domain behavior, and end behavior of a power function.
M39-POW-004	I can recognize and model direct, inverse, and other power-variation relationships.
M39-POW-005	I can solve equations involving integer or rational powers.
M39-POW-006	I can compare the long-run dominance of logarithmic, power, polynomial, and exponential functions.
M39-POW-007	I can fit a power regression model and interpret its parameters and correlation evidence.

Function Combinations and Polynomials

REQUIRED

M39-POL-001	I can add, subtract, multiply, and divide functions represented by tables, graphs, or formulas.
M39-POL-002	I can determine where a combination of functions is positive, zero, or undefined.
M39-POL-003	I can determine polynomial end behavior from degree and leading coefficient.
M39-POL-004	I can identify zeros and multiplicities and relate them to polynomial graph behavior.
M39-POL-005	I can estimate a polynomial's possible degree from its zeros, turning points, inflection points, and end behavior.
M39-POL-006	I can construct a polynomial function from specified zeros, multiplicities, and a point.
M39-POL-007	I can match polynomial formulas and graphs using intercepts, multiplicity, degree, and end behavior.
M39-POL-008	I can build and optimize a polynomial model in a geometric or economic context.

Math 39 Course at a Glance

Precalculus with Data Analysis

Describing Data		Probability		Expected Value		Linearization of Nonlinear Data	
REQUIRED		REQUIRED		EXTENSION		EXTENSION	
M39-STA-001	I can distinguish categorical and quantitative data.	M39-PRO-001	I can organize categorical data in a contingency table or Venn diagram.	M39-EV-001	I can calculate the expected value of a discrete probability situation.	M39-POW-008	I can linearize power or exponential data and translate between the linearized and original models.
M39-STA-002	I can construct and interpret frequency and relative-frequency tables.	M39-PRO-002	I can calculate and interpret empirical, marginal, joint, and conditional probabilities from a contingency table.	M39-EV-002	I can interpret expected value to evaluate a game, risk, or long-run decision.		
M39-STA-003	I can construct and interpret bar charts, pie charts, and histograms using appropriate technology.	M39-PRO-003	I can identify an experiment, outcomes, events, and a sample space.				
M39-STA-004	I can identify misleading features in a data display.	M39-PRO-004	I can calculate theoretical probabilities when outcomes are equally likely.				
M39-STA-005	I can describe a distribution's modality, symmetry, skewness, and outliers.	M39-PRO-005	I can use complements to calculate probabilities.				
M39-STA-006	I can calculate and interpret mean, median, and mode and select an appropriate measure of center.	M39-PRO-006	I can calculate probabilities of compound 'and' events and distinguish independent from dependent events.				
M39-STA-007	I can calculate and interpret range, standard deviation, quartiles, and interquartile range.	M39-PRO-007	I can calculate probabilities of compound 'or' events and distinguish overlapping from disjoint events.				
M39-STA-008	I can construct and interpret a five-number summary and boxplot.	M39-PRO-008	I can compare experimental and theoretical probability and explain the law of large numbers.				
M39-STA-009	I can use percentiles and z-scores to locate and compare values within distributions.						
M39-STA-010	I can compare distributions using their shape, center, spread, and unusual features.						

Math 39 In-Depth Curriculum

Precalculus with Data Analysis

SKILL PROGRESSION

I

Introduce

R

Reinforce

D

Deepen

A

Apply

Linear Modeling and Residuals

REQUIRED

M39-LIN-001	I can calculate and interpret average rate of change with appropriate units.
SUPPORTING SKILLS	
D	Calculate average rate of change over an interval. (first introduced in Math 21) SK-FUNC-AROC · Deepen
D	Interpret average rate of change with contextual units. (first introduced in Math 21) SK-FUNC-AROC-INTERPRET · Deepen
A	Attach and interpret units for rates, inputs, and outputs. (first introduced in Math 12) SK-REP-UNITS · Apply

M39-LIN-002	I can construct a linear function from two data points, a table, a graph, or a context.
SUPPORTING SKILLS	
R	Convert among common forms of a linear equation. (first introduced in Math 12) SK-LIN-FORM-CONVERT · Reinforce
A	Calculate slope from two points. (first introduced in Math 12) SK-LIN-SLOPE-POINTS · Apply
A	Explain a model parameter in the language of its context. (first introduced in Math 21) SK-MODEL-PARAMETER-INTERPRET · Apply

M39-LIN-003	I can fit a linear regression model to data using technology and interpret its parameters.
SUPPORTING SKILLS	
D	Explain a model parameter in the language of its context. (first introduced in Math 21) SK-MODEL-PARAMETER-INTERPRET · Deepen
I	Use technology to determine a least-squares linear regression model. SK-REG-LINEAR-FIT · Introduce

M39-LIN-004	I can interpret a correlation coefficient as evidence about the direction and strength of a linear relationship.
SUPPORTING SKILLS	
I	Interpret the sign and magnitude of a correlation coefficient in context. SK-REG-CORRELATION-INTERPRET · Introduce

M39-LIN-005	I can calculate residuals, construct a residual plot, and use its pattern to evaluate a model.
SUPPORTING SKILLS	
I	Use residual patterns and contextual evidence to judge whether a model is appropriate. SK-REG-MODEL-ASSESS · Introduce
I	Calculate an observed value minus its model-predicted value. SK-REG-RESIDUAL-CALCULATE · Introduce
I	Construct and interpret a residual plot. SK-REG-RESIDUAL-PLOT · Introduce

M39-LIN-006	I can use a linear model to make and interpret predictions, including solving for either variable.
SUPPORTING SKILLS	
A	Identify input and output values in a formula, graph, table, or context. (first introduced in Math 12) SK-FUNC-INPUT-OUTPUT · Apply
D	Solve for an input associated with a specified output. (first introduced in Math 21) SK-FUNC-SOLVE-INPUT · Deepen
A	Restrict a model to meaningful contextual inputs. (first introduced in Math 12) SK-MODEL-VALID-DOMAIN · Apply

Math 39 In-Depth Curriculum

Precalculus with Data Analysis

SKILL PROGRESSION

I Introduce

R Reinforce

D Deepen

A Apply

Exponential Modeling

REQUIRED

M39-EXP-001	I can distinguish linear and exponential change using rates, multipliers, tables, graphs, or contexts.
SUPPORTING SKILLS	
D	Use constant differences or ratios to distinguish linear and exponential patterns. (first introduced in Math 31) SK-EXP-DIFFERENCE-RATIO · Deepen
D	Compare functions shown in different representations. (first introduced in Math 21) SK-FUNC-COMPARE-REP · Deepen

M39-EXP-002	I can construct an exponential function from two data points, a table, or a context.
SUPPORTING SKILLS	
D	Identify initial amount, rate, and time interval in context. (first introduced in Math 31) SK-EXP-MODEL-CONTEXT · Deepen
D	Determine an exponential model from tabular values. (first introduced in Math 31) SK-EXP-MODEL-TABLE · Deepen
D	Solve for an exponential model using two points. (first introduced in Math 31) SK-EXP-MODEL-TWO-POINTS · Deepen

M39-EXP-003	I can interpret the initial value, growth or decay factor, and growth or decay rate of an exponential model.
SUPPORTING SKILLS	
D	Convert a percent increase or decrease to a growth or decay factor. (first introduced in Math 31) SK-EXP-PERCENT-FACTOR · Deepen
D	Explain a model parameter in the language of its context. (first introduced in Math 21) SK-MODEL-PARAMETER-INTERPRET · Deepen

M39-EXP-004	I can determine and interpret doubling time or half-life from an exponential model.
SUPPORTING SKILLS	
A	Identify initial amount, rate, and time interval in context. (first introduced in Math 31) SK-EXP-MODEL-CONTEXT · Apply
D	Translate a doubling-time or half-life question into an exponential equation. (first introduced in Math 31) SK-LOG-DOUBLING-HALF · Deepen

M39-EXP-005	I can solve exponential and logarithmic equations in modeling contexts.
SUPPORTING SKILLS	
A	Reject solutions that make a logarithm argument nonpositive. (first introduced in Math 31) SK-LOG-DOMAIN-CHECK · Apply
R	Isolate an exponential expression and apply a logarithm. (first introduced in Math 31) SK-LOG-SOLVE-EXP · Reinforce
R	Combine or isolate logarithms before converting to exponential form. (first introduced in Math 31) SK-LOG-SOLVE-LOG · Reinforce

M39-EXP-006	I can fit an exponential regression model using technology and evaluate its fit.
SUPPORTING SKILLS	
A	Interpret the sign and magnitude of a correlation coefficient in context. (first introduced in Math 39) SK-REG-CORRELATION-INTERPRET · Apply
I	Use technology to determine an exponential regression model. SK-REG-EXPONENTIAL-FIT · Introduce
D	Use residual patterns and contextual evidence to judge whether a model is appropriate. (first introduced in Math 39) SK-REG-MODEL-ASSESS · Deepen

Math 39 In-Depth Curriculum

Precalculus with Data Analysis

SKILL PROGRESSION

I

Introduce

R

Reinforce

D

Deepen

A

Apply

M39-EXP-007	I can compare linear and exponential models and justify which is more appropriate for a data set or context.
SUPPORTING SKILLS	
A	Compare functions shown in different representations. (first introduced in Math 21) SK-FUNC-COMPARE-REP · Apply
I	Select a function family whose behavior fits a table, graph, or context. SK-MODEL-FAMILY-SELECT · Introduce
D	Use residual patterns and contextual evidence to judge whether a model is appropriate. (first introduced in Math 39) SK-REG-MODEL-ASSESS · Deepen

Math 39 In-Depth Curriculum

Precalculus with Data Analysis

SKILL PROGRESSION



Introduce



Reinforce



Deepen



Apply

Power Functions and Function Comparison

REQUIRED

M39-POW-001	I can identify a power function and interpret the parameters in $y = kx^p$.
SUPPORTING SKILLS	
I	Interpret the coefficient and exponent in a power function. SK-POWER-PARAMETER-INTERPRET · Introduce
I	Represent and interpret a power-variation relationship. SK-VAR-POWER · Introduce

M39-POW-002	I can construct a power function from two data points or a proportional relationship.
SUPPORTING SKILLS	
A	Isolate a specified variable in an equation or formula. (first introduced in Math 12) SK-ALG-VARIABLE-ISOLATE · Apply
I	Determine a power function from two points or a proportional relationship. SK-POWER-CONSTRUCT · Introduce

M39-POW-003	I can relate the sign and size of k and p to the shape, domain behavior, and end behavior of a power function.
SUPPORTING SKILLS	
I	Determine power-function behavior as the input approaches zero or positive or negative infinity. SK-POWER-END-BEHAVIOR · Introduce
I	Relate the coefficient and exponent of a power function to its graph shape. SK-POWER-SHAPE · Introduce
D	Connect parameters in a formula to features of its graph. (first introduced in Math 21) SK-REP-FORMULA-GRAPH · Deepen

M39-POW-004	I can recognize and model direct, inverse, and other power-variation relationships.
SUPPORTING SKILLS	
I	Represent and interpret a direct-variation relationship. SK-VAR-DIRECT · Introduce
I	Represent and interpret an inverse-variation relationship. SK-VAR-INVERSE · Introduce
D	Represent and interpret a power-variation relationship. (first introduced in Math 39) SK-VAR-POWER · Deepen

M39-POW-005	I can solve equations involving integer or rational powers.
SUPPORTING SKILLS	
A	Use inverse operations to maintain an equivalent equation. (first introduced in Math 12) SK-ALG-INVERSE-OPERATIONS · Apply
A	Interpret numerator and denominator roles in a rational exponent. (first introduced in Math 21) SK-EXP-RATIONAL · Apply

M39-POW-006	I can compare the long-run dominance of logarithmic, power, polynomial, and exponential functions.
SUPPORTING SKILLS	
D	Compare functions shown in different representations. (first introduced in Math 21) SK-FUNC-COMPARE-REP · Deepen
I	Compare the long-run growth or decay of functions from different families. SK-FUNC-DOMINANCE · Introduce

Math 39 In-Depth Curriculum

Precalculus with Data Analysis

SKILL PROGRESSION

I

Introduce

R

Reinforce

D

Deepen

A

Apply

M39-POW-007	I can fit a power regression model and interpret its parameters and correlation evidence.
SUPPORTING SKILLS	
D	Interpret the coefficient and exponent in a power function. (first introduced in Math 39) SK-POWER-PARAMETER-INTERPRET · Deepen
A	Interpret the sign and magnitude of a correlation coefficient in context. (first introduced in Math 39) SK-REG-CORRELATION-INTERPRET · Apply
I	Use technology to determine a power regression model. SK-REG-POWER-FIT · Introduce

Math 39 In-Depth Curriculum

Precalculus with Data Analysis

SKILL PROGRESSION



Introduce



Reinforce



Deepen



Apply

Function Combinations and Polynomials

REQUIRED

M39-POL-001	I can add, subtract, multiply, and divide functions represented by tables, graphs, or formulas.
SUPPORTING SKILLS	
A	Identify input and output values in a formula, graph, table, or context. (first introduced in Math 12) SK-FUNC-INPUT-OUTPUT · Apply
I	Add, subtract, multiply, and divide function outputs. SK-FUNC-OPERATIONS · Introduce

M39-POL-002	I can determine where a combination of functions is positive, zero, or undefined.
SUPPORTING SKILLS	
I	Determine the domain of a sum, difference, product, quotient, or composition. SK-FUNC-DOMAIN-COMBINE · Introduce
D	Add, subtract, multiply, and divide function outputs. (first introduced in Math 39) SK-FUNC-OPERATIONS · Deepen

M39-POL-003	I can determine polynomial end behavior from degree and leading coefficient.
SUPPORTING SKILLS	
I	Infer polynomial end behavior from degree and leading coefficient. SK-POLY-END-BEHAVIOR · Introduce
A	Identify terms, factors, coefficients, variables, and degree in a polynomial expression. (first introduced in Math 12) SK-POLY-VOCABULARY · Apply

M39-POL-004	I can identify zeros and multiplicities and relate them to polynomial graph behavior.
SUPPORTING SKILLS	
I	Relate zero multiplicity to local graph behavior. SK-POLY-MULTIPLICITY · Introduce
D	Determine real and complex zeros using an appropriate method. (first introduced in Math 21) SK-POLY-ZEROS · Deepen

M39-POL-005	I can estimate a polynomial's possible degree from its zeros, turning points, inflection points, and end behavior.
SUPPORTING SKILLS	
D	Infer polynomial end behavior from degree and leading coefficient. (first introduced in Math 39) SK-POLY-END-BEHAVIOR · Deepen
D	Relate zero multiplicity to local graph behavior. (first introduced in Math 39) SK-POLY-MULTIPLICITY · Deepen

M39-POL-006	I can construct a polynomial function from specified zeros, multiplicities, and a point.
SUPPORTING SKILLS	
D	Build a polynomial from zeros, multiplicities, and a point. (first introduced in Math 31) SK-POLY-CONSTRUCT · Deepen
A	Relate zero multiplicity to local graph behavior. (first introduced in Math 39) SK-POLY-MULTIPLICITY · Apply
A	Determine real and complex zeros using an appropriate method. (first introduced in Math 21) SK-POLY-ZEROS · Apply

Math 39 In-Depth Curriculum

Precalculus with Data Analysis

SKILL PROGRESSION

I

Introduce

R

Reinforce

D

Deepen

A

Apply

M39-POL-007	I can match polynomial formulas and graphs using intercepts, multiplicity, degree, and end behavior.
SUPPORTING SKILLS	
D	Infer polynomial end behavior from degree and leading coefficient. (first introduced in Math 39) SK-POLY-END-BEHAVIOR · Deepen
A	Relate zero multiplicity to local graph behavior. (first introduced in Math 39) SK-POLY-MULTIPLICITY · Apply
A	Determine real and complex zeros using an appropriate method. (first introduced in Math 21) SK-POLY-ZEROS · Apply

M39-POL-008	I can build and optimize a polynomial model in a geometric or economic context.
SUPPORTING SKILLS	
A	Explain a model parameter in the language of its context. (first introduced in Math 21) SK-MODEL-PARAMETER-INTERPRET · Apply
A	Restrict a model to meaningful contextual inputs. (first introduced in Math 12) SK-MODEL-VALID-DOMAIN · Apply
D	Interpret vertex and zeros in an application. (first introduced in Math 21) SK-QUAD-MODEL-FEATURES · Deepen

Math 39 In-Depth Curriculum

Precalculus with Data Analysis

SKILL PROGRESSION



Introduce



Reinforce



Deepen



Apply

Describing Data

REQUIRED

M39-STA-001	I can distinguish categorical and quantitative data.
SUPPORTING SKILLS	
I	Classify data as categorical or quantitative. SK-STAT-DATA-TYPE · Introduce

M39-STA-002	I can construct and interpret frequency and relative-frequency tables.
SUPPORTING SKILLS	
I	Construct and interpret a frequency table. SK-STAT-FREQUENCY-TABLE · Introduce
I	Calculate and interpret relative frequency. SK-STAT-RELATIVE-FREQUENCY · Introduce

M39-STA-003	I can construct and interpret bar charts, pie charts, and histograms using appropriate technology.
SUPPORTING SKILLS	
I	Construct and interpret bar charts and pie charts. SK-STAT-CATEGORICAL-DISPLAY · Introduce
I	Construct and interpret a histogram for quantitative data. SK-STAT-HISTOGRAM · Introduce

M39-STA-004	I can identify misleading features in a data display.
SUPPORTING SKILLS	
I	Identify perceptual distortion, misleading scales, and other misleading features in data displays. SK-STAT-GRAPH-CRITIQUE · Introduce

M39-STA-005	I can describe a distribution's modality, symmetry, skewness, and outliers.
SUPPORTING SKILLS	
I	Describe a distribution by modality, symmetry, and skewness. SK-STAT-DISTRIBUTION-SHAPE · Introduce
I	Identify and interpret unusual observations or outliers. SK-STAT-OUTLIER-IDENTIFY · Introduce

M39-STA-006	I can calculate and interpret mean, median, and mode and select an appropriate measure of center.
SUPPORTING SKILLS	
I	Select an appropriate measure of center based on distribution shape and context. SK-STAT-CENTER-SELECT · Introduce
I	Calculate and interpret the arithmetic mean. SK-STAT-MEAN · Introduce
I	Calculate and interpret the median. SK-STAT-MEDIAN · Introduce
I	Identify and interpret the mode or modes. SK-STAT-MODE · Introduce

M39-STA-007	I can calculate and interpret range, standard deviation, quartiles, and interquartile range.
SUPPORTING SKILLS	
I	Determine quartiles and calculate and interpret the interquartile range. SK-STAT-QUARTILES-IQR · Introduce
I	Calculate and interpret the range. SK-STAT-RANGE · Introduce
I	Calculate and interpret standard deviation as a measure of spread. SK-STAT-STANDARD-DEVIATION · Introduce

Math 39 In-Depth Curriculum

Precalculus with Data Analysis

SKILL PROGRESSION

I

Introduce

R

Reinforce

D

Deepen

A

Apply

M39-STA-008	I can construct and interpret a five-number summary and boxplot.
SUPPORTING SKILLS	
I	Construct and interpret a boxplot. SK-STAT-BOXPLOT · Introduce
I	Determine and interpret a five-number summary. SK-STAT-FIVE-NUMBER · Introduce

M39-STA-009	I can use percentiles and z-scores to locate and compare values within distributions.
SUPPORTING SKILLS	
I	Determine and interpret the percentile rank of a value. SK-STAT-PERCENTILE · Introduce
I	Calculate and interpret a z-score. SK-STAT-ZSCORE · Introduce

M39-STA-010	I can compare distributions using their shape, center, spread, and unusual features.
SUPPORTING SKILLS	
A	Select an appropriate measure of center based on distribution shape and context. (first introduced in Math 39) SK-STAT-CENTER-SELECT · Apply
I	Compare distributions using shape, center, spread, and unusual features. SK-STAT-DISTRIBUTION-COMPARE · Introduce
A	Describe a distribution by modality, symmetry, and skewness. (first introduced in Math 39) SK-STAT-DISTRIBUTION-SHAPE · Apply

Math 39 In-Depth Curriculum

Precalculus with Data Analysis

SKILL PROGRESSION

I Introduce

R Reinforce

D Deepen

A Apply

Probability

REQUIRED

M39-PRO-001

I can organize categorical data in a contingency table or Venn diagram.

SUPPORTING SKILLS

I

Construct and interpret a contingency table.
SK-PROB-CONTINGENCY-TABLE · Introduce

I

Represent categorical events and their overlap with a Venn diagram.
SK-PROB-VENN-DIAGRAM · Introduce

M39-PRO-002

I can calculate and interpret empirical, marginal, joint, and conditional probabilities from a contingency table.

SUPPORTING SKILLS

I

Calculate and interpret a conditional probability.
SK-PROB-CONDITIONAL · Introduce

I

Calculate probability from observed relative frequency.
SK-PROB-EMPIRICAL · Introduce

I

Calculate and interpret marginal and joint probabilities.
SK-PROB-MARGINAL-JOINT · Introduce

M39-PRO-003

I can identify an experiment, outcomes, events, and a sample space.

SUPPORTING SKILLS

I

Identify outcomes, events, and the sample space of an experiment.
SK-PROB-SAMPLE-SPACE · Introduce

M39-PRO-004

I can calculate theoretical probabilities when outcomes are equally likely.

SUPPORTING SKILLS

A

Identify outcomes, events, and the sample space of an experiment. (first introduced in Math 39)
SK-PROB-SAMPLE-SPACE · Apply

I

Calculate theoretical probability for equally likely outcomes.
SK-PROB-THEORETICAL · Introduce

M39-PRO-005

I can use complements to calculate probabilities.

SUPPORTING SKILLS

I

Use the complement rule to calculate probability.
SK-PROB-COMPLEMENT · Introduce

M39-PRO-006

I can calculate probabilities of compound 'and' events and distinguish independent from dependent events.

SUPPORTING SKILLS

I

Calculate the probability of a compound and event.
SK-PROB-AND · Introduce

D

Calculate and interpret a conditional probability. (first introduced in Math 39)
SK-PROB-CONDITIONAL · Deepen

I

Determine whether events are independent or dependent.
SK-PROB-INDEPENDENCE · Introduce

M39-PRO-007

I can calculate probabilities of compound 'or' events and distinguish overlapping from disjoint events.

SUPPORTING SKILLS

I

Determine whether events are overlapping or disjoint.
SK-PROB-DISJOINT · Introduce

I

Calculate the probability of a compound or event.
SK-PROB-OR · Introduce

Math 39 In-Depth Curriculum

Precalculus with Data Analysis

SKILL PROGRESSION

I

Introduce

R

Reinforce

D

Deepen

A

Apply

M39-PRO-008	I can compare experimental and theoretical probability and explain the law of large numbers.
SUPPORTING SKILLS	
D	Calculate probability from observed relative frequency. (first introduced in Math 39) SK-PROB-EMPIRICAL · Deepen
I	Explain how experimental probability stabilizes with many trials. SK-PROB-LARGE-NUMBERS · Introduce
D	Calculate theoretical probability for equally likely outcomes. (first introduced in Math 39) SK-PROB-THEORETICAL · Deepen

Math 39 In-Depth Curriculum

Precalculus with Data Analysis

SKILL PROGRESSION

I

Introduce

R

Reinforce

D

Deepen

A

Apply

Expected Value

EXTENSION

M39-EV-001	I can calculate the expected value of a discrete probability situation.
SUPPORTING SKILLS	
I	Calculate the expected value of a discrete probability situation. SK-PROB-EXPECTED-VALUE · Introduce
M39-EV-002	I can interpret expected value to evaluate a game, risk, or long-run decision.
SUPPORTING SKILLS	
I	Interpret expected value as a long-run average for decision-making. SK-PROB-EXPECTED-INTERPRET · Introduce
D	Calculate the expected value of a discrete probability situation. (first introduced in Math 39) SK-PROB-EXPECTED-VALUE · Deepen

Math 39 In-Depth Curriculum

Precalculus with Data Analysis

SKILL PROGRESSION

I

Introduce

R

Reinforce

D

Deepen

A

Apply

Linearization of Nonlinear Data

EXTENSION

M39-POW-008	I can linearize power or exponential data and translate between the linearized and original models.
SUPPORTING SKILLS	
I	Transform exponential data so that a linear model can be fit to its logarithms. SK-REG-LOG-LINEARIZE · Introduce
I	Transform power data so that a linear model can be fit on logarithmic axes. SK-REG-LOGLOG-LINEARIZE · Introduce
I	Translate a linearized regression equation back to its original nonlinear form. SK-REG-UNLINEARIZE · Introduce